

Worksheet 7B: Linear Systems and Gaussian Elimination

Series 7: Linear Algebra
AIML Foundations Mathematics
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What This Worksheet Is About

In ws07a you learned that a system of equations can be written $A\mathbf{x} = \mathbf{b}$, and solved via $A^{-1}\mathbf{b}$. Gaussian elimination is the systematic, efficient method computers actually use. It reveals a deeper truth: not every system has a unique solution — some have none, some have infinitely many. Understanding which case you are in is fundamental to data fitting and numerical computing.

Part A: Augmented Matrices (8 problems)

A system is encoded as an augmented matrix $[A \mid \mathbf{b}]$: $\begin{cases} 2x + 3y = 7 \\ x - y = 1 \end{cases} \longrightarrow \left[\begin{array}{cc|c} 2 & 3 & 7 \\ 1 & -1 & 1 \end{array} \right]$

1. Write $3x - 2y = 5$, $x + 4y = -3$ as an augmented matrix.
2. Write $x + y + z = 6$, $2x - y + 3z = 11$, $x + 2y - z = 2$ as an augmented matrix.
3. Write $4x = 12$, $x + y = 7$, $y - z = 1$ as an augmented matrix.
4. Write $2x_1 - x_2 + 3x_3 - x_4 = 4$, $x_1 + 2x_2 - x_3 + x_4 = 1$ as an augmented matrix.
5. Read the system from $\left[\begin{array}{cc|c} 1 & -3 & 5 \\ 0 & 1 & -2 \end{array} \right]$ and solve by back substitution.
6. Read the system from $\left[\begin{array}{ccc|c} 1 & 0 & 2 & 4 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & 2 \end{array} \right]$ and solve.
7. Read the solution directly from $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 3 \end{array} \right]$. Why is this form useful?
8. Write the augmented matrix for: $w_1 + 2w_2 + w_3 = 5$, $2w_1 - w_2 + 3w_3 = 8$, $w_1 + w_2 - w_3 = 1$.

Part B: Row Operations (10 problems)

Three legal row operations (all reversible, all preserve the solution set):

- R1. Swap rows: $R_i \leftrightarrow R_j$
R2. Scale a row: $kR_i \rightarrow R_i$ ($k \neq 0$)

R3. Replace: $R_i + kR_j \rightarrow R_i$

Goal: reach **row echelon form (REF)** — a staircase of leading entries (pivots) with zeros below.

1. Apply $\frac{1}{2}R_1 \rightarrow R_1$ to $\left[\begin{array}{cc|c} 2 & 4 & 6 \\ 3 & -1 & 5 \end{array} \right]$.

2. Apply $R_2 - 3R_1 \rightarrow R_2$ to $\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 3 & -1 & 5 \end{array} \right]$.

3. Apply $R_3 - 2R_1 \rightarrow R_3$ to $\left[\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 0 & 3 & 2 & 7 \\ 2 & -1 & 3 & 1 \end{array} \right]$.

4. Reduce to REF: $\left[\begin{array}{cc|c} 2 & 6 & 10 \\ 1 & 2 & 4 \end{array} \right]$

5. Reduce to REF: $\left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 2 & 1 & -1 & 1 \\ 1 & -1 & 2 & 3 \end{array} \right]$

6. Reduce to REF (swap rows first): $\left[\begin{array}{ccc|c} 0 & 1 & 2 & 5 \\ 1 & 3 & -1 & 2 \\ 2 & 0 & 1 & 3 \end{array} \right]$

7. Which are in row echelon form? (a) $\left[\begin{array}{ccc} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{array} \right]$ (b) $\left[\begin{array}{ccc} 1 & 0 & 2 \\ 0 & 0 & 1 \\ 0 & 3 & 4 \end{array} \right]$ (c) $\left[\begin{array}{cc} 2 & 1 \\ 0 & 3 \\ 0 & 0 \end{array} \right]$ (d) $\left[\begin{array}{ccc} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{array} \right]$

8. Show each row operation is reversible. Why does reversibility guarantee the solution set is unchanged?

9. Why is dividing a row by zero not permitted? What must be true about pivots?

10. Why is $i = j$ forbidden in $R_i + kR_j \rightarrow R_i$?

Part C: Back Substitution and Gauss–Jordan (10 problems)

Back substitution: once in REF, solve from the bottom row upward.

Gauss–Jordan: continue to **reduced row echelon form (RREF)**: each pivot is 1 and all entries above and below are 0.

11. Solve by elimination + back substitution: $x + 2y = 5$, $3x - y = 4$.

12. Solve: $2x + y - z = 3$, $x - y + 2z = 1$, $3x + 2y + z = 8$.

13. Solve: $x_1 + x_2 + x_3 = 6$, $2x_1 - x_2 + 3x_3 = 11$, $x_1 + 2x_2 - x_3 = 2$.

14. Use Gauss–Jordan (RREF) to solve: $3x + 6y = 12$, $x + 2y - z = 4$, $2x + 4y + z = 10$.

15. Use Gauss–Jordan to solve: $x + y + z = 6$, $x - y + z = 2$, $2x + y - z = 3$.

16. Find A^{-1} via row-reducing $[A \mid I]$ for $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$.
17. Explain why row-reducing $[A \mid I] \rightarrow [I \mid A^{-1}]$ gives the inverse.
18. Solve problem 19 again using A^{-1} from problem 24. Do you get the same answer?
19. For the RREF $\left[\begin{array}{ccc|c} 1 & 0 & 2 & 3 \\ 0 & 1 & -1 & 1 \end{array} \right]$, identify the pivot variables and free variable. Write the general solution.
20. If A is 3×3 and you row-reduce $[A \mid I]$ but the left side never reaches I (a zero row appears), what does this tell you about A ?

Part D: Types of Solutions (10 problems)

Three possibilities after row reduction:

1. **Unique solution:** n pivots, n unknowns, consistent.
 2. **No solution (inconsistent):** row $[0 \cdots 0 \mid c]$ with $c \neq 0$.
 3. **Infinitely many:** fewer pivots than unknowns; free variables exist.
21. Classify and solve (or show no solution): $x + 2y = 3$, $2x + 4y = 6$.
 22. Classify and solve: $x + 2y = 3$, $2x + 4y = 7$.
 23. Classify and solve: $x - y + z = 2$, $2x - 2y + 2z = 5$.
 24. Classify and solve: $x + y + z = 6$, $x - y + 2z = 5$, $2x + 3z = 11$.
 25. Classify and solve: $x_1 + 2x_2 - x_3 + x_4 = 3$, $2x_1 + 4x_2 + x_3 - x_4 = 7$, $x_1 + 2x_2 + 2x_3 - 2x_4 = 4$.
 26. For a system with 3 equations and 4 unknowns, why is a unique solution impossible?
 27. A homogeneous system $A\mathbf{x} = \mathbf{0}$ always has $\mathbf{x} = \mathbf{0}$. Under what condition does it have infinitely many solutions?
 28. Find the rank of: (a) $\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$
 29. Find the null space of $A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \end{bmatrix}$.
 30. Which vectors $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ are in the column space of the matrix in problem 34?

Part E: Applications (6 problems)

31. **Circuit.** $I_1 - I_2 - I_3 = 0$, $2I_1 + 3I_2 = 12$, $3I_2 - 2I_3 = 6$. Solve for I_1, I_2, I_3 .
32. **Least Squares.** For $A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$: (a) Compute $A^T A$ and $A^T \mathbf{b}$. (b) Solve the normal equations $A^T A \mathbf{x} = A^T \mathbf{b}$. (c) Interpret the result as a line of best fit.

33. In problem 37, why can't we compute $A^{-1}\mathbf{b}$ directly? What makes this an over-determined system?
34. **Chemistry.** Balance $a\text{C}_2\text{H}_6 + b\text{O}_2 \rightarrow c\text{CO}_2 + d\text{H}_2\text{O}$. Write and solve the linear system (set $a = 1$).
35. **Network flow.** At 4 intersections, flow must balance: in = out. Flows: into node 1 = $50 + x_4$, out = $x_1 + x_2$; into node 2 = x_1 , out = $x_3 + 30$; into node 3 = $x_2 + x_3$, out = $40 + x_4$; into node 4 = external 20, out = x_4 . Write the augmented matrix and solve.
36. Reflection: (a) Why is Gaussian elimination preferred over computing A^{-1} for large systems? (b) The three solution types correspond geometrically to what? (c) What happens in least squares when $A^T A$ is not invertible?